구배 굴절률 메타물질을 이용한 음파 조작

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Acoustic Wave Manipulation Using Gradient Refractive Index Metamaterial

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Abstract

In this study, an Eaton lens, a type of refractive index (GRIN) lens, was implemented as a two-dimensional acoustic metamaterial. This Eaton lens can bend the incident wave in a desired direction, and a lens with a bending angle of 90° was studied here. By arranging meta-atoms in unit cells inside the lens divided into several layers, the density of the space through which acoustic waves pass is controlled, resulting in acoustic metamaterials that satisfy the refractive index of Eaton lenses. has been implemented A meta-lens with such a beam steering function is expected to be applied as an underwater acoustic antenna or seismic wave control.

I. 서론

Acoustic metamaterials can control sound waves. The refractive index (n) of these materials is controlled by the mass density ρ and bulk modulus B.

$$n = \sqrt{\frac{\rho}{B}} \tag{1}$$

Eaton lens proposed in 1952 [1] as part of the transformation optics can also be realized as an acoustic metamaterial. The refractive index formula is as follows [2]:

$$n^{\frac{\pi}{\theta}} = \frac{R}{nr} + \sqrt{\left(\frac{R}{nr}\right)^2 - 1} \tag{2}$$

where R is the radius of lens, r is the radius of each layer, θ is the bending angle, and n is the refractive index. The refractive index solution is not analytical except for a 90°, 180°, and 360°. Therefore, it can be calculated by numerical simulations.

To solve this problem, the generalized refractive index was derived as shown in Eq. (3).[3,4]

$$n(r,\theta) \simeq \left(\frac{2R}{r} - 1\right)^{\frac{\theta}{\pi+\theta}} \tag{3}$$

In this paper, 90° (right bender) was realized as an acoustic metamaterial, rather than the previously reported 15°, 45°, [5] and 180° (retroreflector) [6]. analytical and numerical models were compared through simulations. As a result, these acoustic metamaterials are expected as promising applications for controlling waves.

II. 설계

The lens is discretized into N layers, consisting of several unit cells with a side length b and meta−atoms with a radius a_i, and the formula is as follows:

$$n(i,\theta) \simeq \left(\frac{2N}{i}-1\right)^{\frac{\theta}{\pi+\theta}} \qquad (4)$$

where i (natural number) is the order of the layers. The design method by calculating the dimension of the structure constituting the lens using Eq. (4) and lens are shown in Fig. 1(a) and Fig. 1(b), respectively.

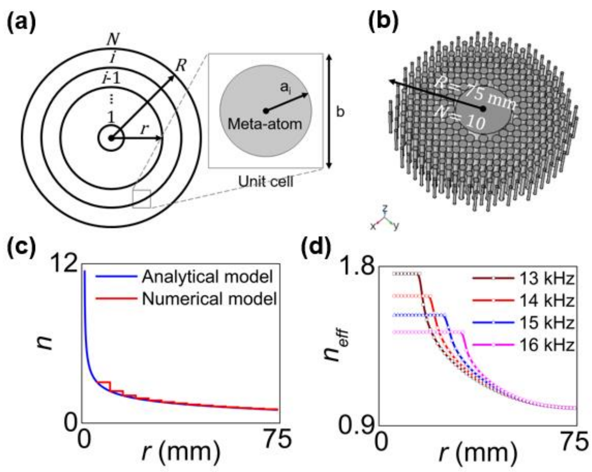

그림 1. Design of the lens and its refractive index. (a) is a schematic diagram of the design method divided into several layers, and (b) is the acoustic Eaton meta−lens with a radius of 75 mm and consisting of 10 layers. (c) and (d) are the refractive for the analytical and numerical models and effective refractive index, respectively.

Fig. 1(c) shows the refractive index of the analytical and numerical models using Eq. (3) and Eq. (4). In the center of the lens, there is a singularity in which the refractive index is infinite. It means that the meta−atoms are larger than the unit cell. Therefore, it is made up of a large cylinder up to the third layer of the numerical lens, as shown in Fig. (2).

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Eq. (5) was used to calculate the effective refractive index (n_{eff}) of the lens. [7]

$$n_{eff} = \frac{1}{kd}\left\{\pm\cos^{-1}\left(\frac{1}{2T}\left[1-\left(R^2-T^2\right)\right]\right)+2\pi m\right\} \qquad (5)$$

where k is the wavenumber, d is the slab thickness, and m is the integer number. It is obtained by calculating the reflection (R) and transmission (T) coefficients using finite element method simulation, as shown in Fig. 1(d). The effective refractive index depends on the frequency applied to the lens. The acoustic metamaterials also have different operating ranges depending on the frequency.

III. 결과

First, a transient analysis of the analytical model was performed. It was simulated in the air surrounded perfectly matched layer (PML), and Fig. 2 shows the trajectory of the sound waves as it enters the lens in chronological order. As a result, it was confirmed that the sound waves bend about 90° from the center of the lens.

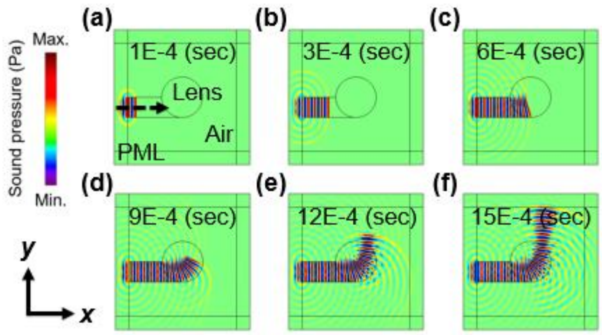

그림 2. Simulation results of the transient analysis. These are the results over time from (a) to (f).

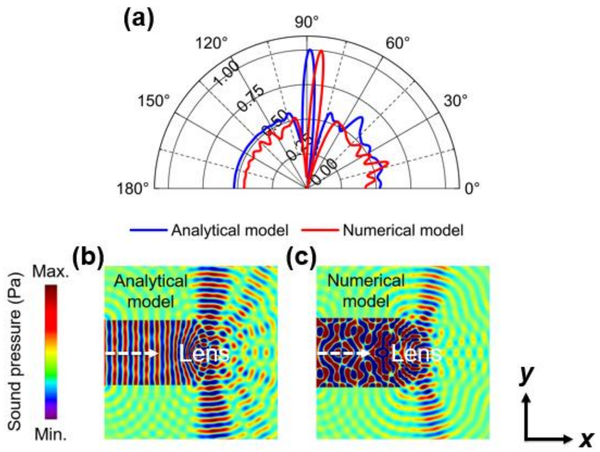

그림 3. Simulation results of the analytical and numerical model. (a) is the beam pattern graph for the analytical and numerical models. The values of the sound pressure were normalized from 0 to 1. (b) and (c) are the sound pressure field (Pa) for the both models. The applied frequency is 14 kHz.

Furthermore, Fig. 3(b) and Fig. 3(c) show the sound pressure fields (Pa) for both models. In the case of the numerical model, it was confirmed that the analytical refractive index could not be satisfied by singularity and reflected sound waves, so it bends slightly less than 90°.

Ⅳ. 결론

Acoustic Eaton meta-lens with a bending angle of 90° was numerically calculated. Because of its singularity, a large cylinder column is placed in the center of the lens. By controlling the wave in the desired direction, the acoustic metamaterials will be suitable for applications such as underwater acoustic antennas and seismic lenses in the future.

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